

DBMS ASSIGNMENT

Database Design for a College Management System

Real-World Application: College Academic and Student Management Platform

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1. Name and Purpose of the Application

Application Name: College Management System

Purpose: A College Management System is used to manage the academic and administrative information of a college. Students can be registered under departments, faculty members can be assigned to courses, students can enroll in courses, and their attendance, examinations, and results can be maintained.

The main problem this application solves is keeping student and academic records organized in one database. Instead of maintaining separate manual records, the college can use connected tables to track which students belong to which department, which courses they take, who teaches those courses, and how students perform in examinations.

2. Required Database Tables

Eight tables are enough to cover the core academic management process. These tables handle students, departments, faculty, courses, enrollment, attendance, examinations, and results.

1. Departments
2. Students
3. Faculty
4. Courses
5. Enrollments
6. Attendance
7. Examinations
8. Results

3. Columns / Attributes for Each Table

Table 1: Departments

Column	Data Type	Key	Purpose
department_id	INT	PK	Unique ID for each department
department_name	VARCHAR		Name of the department
hod_name	VARCHAR		Name of the Head of Department
office_email	VARCHAR	UNIQUE	Department email address

Table 2: Students

Column	Data Type	Key	Purpose
student_id	INT	PK	Unique ID for each student
student_name	VARCHAR		Full name of the student
date_of_birth	DATE		Student's date of birth
gender	VARCHAR		Gender of the student
email	VARCHAR	UNIQUE	Student email address

Column	Data Type	Key	Purpose
phone	VARCHAR		Student contact number
department_id	INT	FK	Department to which the student belongs
admission_year	INT		Year in which the student was admitted

Table 3: Faculty

Column	Data Type	Key	Purpose
faculty_id	INT	PK	Unique ID for each faculty member
faculty_name	VARCHAR		Full name of faculty member
email	VARCHAR	UNIQUE	Faculty email address
phone	VARCHAR		Faculty contact number
designation	VARCHAR		Position such as Professor or Assistant Professor
department_id	INT	FK	Department where the faculty member works

Table 4: Courses

Column	Data Type	Key	Purpose
course_id	INT	PK	Unique ID for each course
course_name	VARCHAR		Name of the course or subject
credits	INT		Number of credits assigned to the course
semester	INT		Semester in which the course is offered
department_id	INT	FK	Department offering the course
faculty_id	INT	FK	Faculty member teaching the course

Table 5: Enrollments

Column	Data Type	Key	Purpose
enrollment_id	INT	PK	Unique ID for each enrollment
student_id	INT	FK	Student who is enrolled
course_id	INT	FK	Course selected by the student
enrollment_date	DATE		Date of enrollment
academic_year	VARCHAR		Academic year of enrollment

Table 6: Attendance

Column	Data Type	Key	Purpose
attendance_id	INT	PK	Unique attendance record ID
student_id	INT	FK	Student whose attendance is recorded
course_id	INT	FK	Course for which attendance is recorded
total_classes	INT		Total classes conducted
classes_attended	INT		Number of classes attended
attendance_percentage	DECIMAL		Student's attendance percentage

Table 7: Examinations

Column	Data Type	Key	Purpose
exam_id	INT	PK	Unique examination ID
course_id	INT	FK	Course for which the exam is conducted
exam_name	VARCHAR		Name/type of examination
exam_date	DATE		Date of examination
max_marks	INT		Maximum marks for the examination

Table 8: Results

Column	Data Type	Key	Purpose
result_id	INT	PK	Unique result ID
student_id	INT	FK	Student receiving the result
exam_id	INT	FK	Examination for which the result is recorded
marks_obtained	DECIMAL		Marks scored by the student
grade	VARCHAR		Grade obtained
result_status	VARCHAR		Pass or Fail status

4. Relationships Between the Tables

Tables	Relationship	Explanation
Departments → Students	One-to-Many	One department can have many students, but each student belongs to one department.
Departments → Faculty	One-to-Many	One department can have several faculty members, while each faculty member belongs to one department.
Departments → Courses	One-to-Many	A department can offer multiple courses, while each course belongs to one department.
Faculty → Courses	One-to-Many	One faculty member can teach multiple courses, while each course is assigned to a faculty member.
Students → Enrollments	One-to-Many	One student can have several enrollment records.
Courses → Enrollments	One-to-Many	One course can have many enrolled students.
Students → Attendance	One-to-Many	A student can have attendance records for multiple courses.
Courses → Attendance	One-to-Many	A course can have attendance records for many students.
Courses → Examinations	One-to-Many	A course can have multiple examinations.
Students → Results	One-to-Many	A student can have multiple examination results.
Examinations → Results	One-to-Many	One examination can produce results for many students.

The **Students and Courses** relationship is effectively **many-to-many**, because one student can study multiple courses and one course can have multiple students. The **Enrollments** table is used to manage this relationship.

Simplified ER Relationship

Departments 1 ■■< Students

Departments 1 ■■< Faculty

Departments 1 ■■< Courses

Faculty 1 ■■< Courses

Students 1 ■■< Enrollments >■■ 1 Courses

Students 1 ■■< Attendance >■■ 1 Courses

Courses 1 ■■< Examinations

Students 1 ■■< Results >■■ 1 Examinations

5. Brief Purpose of Each Table

Table	Purpose
Departments	Stores information about the different departments in the college.
Students	Holds personal and academic details of students and identifies their department.
Faculty	Stores information about teaching staff and their departments.
Courses	Maintains the subjects offered by different departments and the faculty assigned to teach them.
Enrollments	Records which students are enrolled in which courses.
Attendance	Keeps track of classes conducted, classes attended, and attendance percentage for students.
Examinations	Stores information about examinations conducted for different courses.
Results	Records marks, grades, and pass/fail status obtained by students in examinations.

6. Conclusion

A College Management System can be effectively represented using these eight interconnected tables. The **Students, Courses, and Enrollments** tables form the core of the academic structure, while **Attendance, Examinations, and Results** maintain student performance records. **Departments and Faculty** provide the organizational structure required to manage courses and teaching staff.

Primary keys ensure that every record is uniquely identified, while foreign keys connect related records between tables. Overall, this database design provides a practical way to manage student information, course enrollment, attendance, examinations, and results without maintaining unnecessary or unrelated data.